

EuroSAT RGB: Baseline and Error Investigation

Part A — Reproducible baseline

I built a 10-class EuroSAT RGB image-classification pipeline using ResNet-18. With seed 42, the split was 18,900 training, 4,050 validation and 4,050 test images (70/15/15). Best validation accuracy was 96.44%; held-out test accuracy was 97.06%.

Test accuracy	Best validation	Split sizes (train/val/test)	Seed
97.06%	96.44%	18,900 / 4,050 / 4,050	42

Part B

Hypothesis (H1): Are farmland classification errors primarily caused by label ambiguity?

H1: The model's farmland errors are mostly caused by ambiguous or mislabelled tiles, rather than by model weakness.

After the baseline reached 97.1% test accuracy, I looked at the 119 mistakes instead of the aggregate number. They were concentrated, 54 were predicted as AnnualCrop, and Pasture and PermanentCrop together accounted for 57 of the errors. The most confident mistakes also included tiles that looked mislabelled or ambiguous to me, such as a uniformly dark tile labelled AnnualCrop that the model called SeaLake. That suggested the errors might come from the data rather than the model. I chose to test this because it makes a prediction that can fail (ambiguity should be higher in mistakes than in correct predictions), it needs only the saved predictions, so it is reproducible without retraining.

The hypothesis was tested in two stages.

Stage 1 — Initial error review (54 images predicted as AnnualCrop)

- Prediction: At least 30% of the misclassified images would be labelled label_wrong or ambiguous.
- Result: 24 of 54 images (44.4%) were classified as unclear.
- Initial interpretation: This appeared to support H1 because a substantial proportion of errors involved potentially ambiguous or incorrect labels.

Stage 2 — Control experiment (60 Pasture and PermanentCrop images)

- Prediction: If label ambiguity contributes substantially to model errors, unclear labels should be noticeably more common among misclassified images than among correctly classified images.
- Result: 14 of 30 misclassified images (46.7%) and 13 of 30 correctly classified images (43.3%) were unclear—a difference of only 3.4 percentage points.
- Interpretation: The similar rates indicate that unclear labels were also common among correctly classified images. Therefore, the initial review did not establish that label ambiguity was responsible for the model's errors.

Method and findings

I reviewed 60 images: 30 model errors and 30 correct predictions. For each, I recorded one of three judgments: clear, ambiguous, or label appears wrong. For the comparison, ambiguous and label-wrong cases were grouped as “unclear.”

Sample	Unclear cases	Unclear rate
Model errors (n=30)	14 / 30	46.7%
Correct predictions (n=30)	13 / 30	43.3%
Difference	—	3.4 percentage points

The unclear-label rates were nearly the same for errors and correct predictions (46.7% versus 43.3%). In this sample, unclear labels did not distinguish model failures from successful predictions. This weakens the hypothesis that label ambiguity is the main explanation for the observed errors; it suggests I should investigate model/data behavior further rather than attribute errors to labels based on a few examples.

Limitations and next experiment

The review covered only two classes and a small sample of images, so the results may not represent the full dataset. Since the images were judged manually by one person, some decisions may be subjective. As a next step, I would check image similarity and complete the duplicate test ie, remove the most similar training images, and compare the results with random image removal while keeping the evaluation setup the same